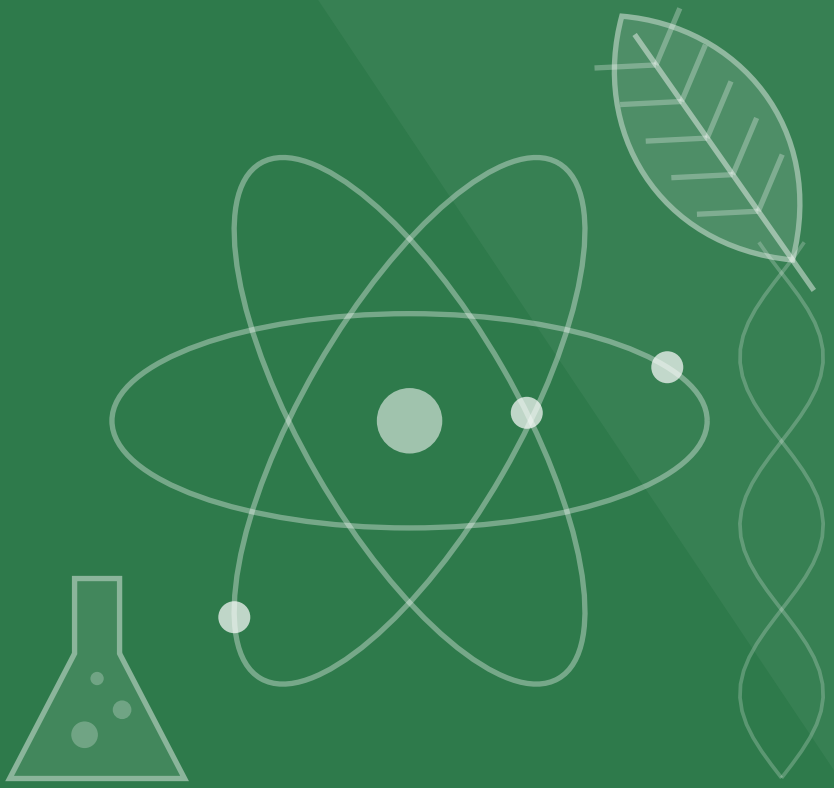
VENVLEARNING

STUDENT TEXTBOOK



Science

GRADE 7

Lessons, examples, and practice for the whole year

2 units

A VenvLearning family homeschool curriculum

GRADE 7 · SCIENCE

VENVLEARNING

Grade 7
Science

STUDENT TEXTBOOK

Unit 1 Cycles of Matter and Energy

Unit 2 Properties and Changes of Matter

A VenvLearning family homeschool curriculum

How to use this book

Every lesson starts on a new page with the same parts, so you always know where to look. Read the Explain It boxes to see how a scientist reasons from evidence, and answer the Check Your Understanding questions. Your practice pages are in your Student Workbook, labeled with the same lesson number.

- Today's Goal (the box with the target) says what you will be able to do by the end of the lesson.
- Today's Assignment is your checklist for the day. Tick each box as you finish it.
- Each lesson page has three panels. The wide center panel is the lesson, in numbered steps: 1 Get Ready, 2 Learn, and 3 Check Your Understanding.
- The left panel holds things to remember: Key Ideas (the most important rule or fact; learn these), Watch Out warnings about common mistakes, tips, and a Remember note that sums up the lesson. Each one sits beside the part of the lesson it goes with.
- The right panel holds Words to Know (definitions), what to read in your Reader, and other extras.
- Pages that look like a page from a book, on cream paper with a brown READING tab, are things to read: stories, excerpts, and articles.
- Investigate boxes are hands-on experiments. Do them with your teacher and record what you observe.
- At the end of each week of lessons there is a short Weekly Check in your workbook. Each unit ends with the Unit Test.

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UNIT 1

Cycles of Matter and Energy

Science practices, Earth's spheres, the water, carbon, oxygen, and nitrogen cycles, photosynthesis, respiration, food webs, and biodiversity.

Every atom in your body has been somewhere else before: in the air, the ocean, a plant, or another animal. In this unit you'll learn how scientists design experiments and explain data, then use those skills to trace how matter cycles and energy flows through Earth's systems and living things.

In this unit you will…

- Identify variables and write CER explanations from data.
- Model the water, carbon, oxygen, and nitrogen cycles.
- Explain photosynthesis and cellular respiration as linked processes.
- Model energy flow in food webs and evaluate ways to protect biodiversity.

SCIENCE IN YOUR WORLD

Farmers, foresters, and environmental engineers use these cycles to grow food, manage forests, and protect water supplies.

UNIT VOCABULARY

independent variable	The variable that is changed.
dependent variable	The variable that is measured.
photosynthesis	Light + CO_2 + water → glucose + oxygen.
cellular respiration	Glucose + oxygen → CO_2 + water + energy.
nitrogen fixation	Changing N_2 into usable forms.
biodiversity	The variety of life in an area.

Lessons in this unit

Lesson	Title
1.1	Variables, Data, and Scientific Explanations
1.2	Earth's Spheres and the Water Cycle
1.3	The Carbon and Oxygen Cycles
1.4	The Nitrogen Cycle
1.5	Photosynthesis
1.6	Cellular Respiration
1.7	Food Webs and Energy Pyramids
1.8	Biodiversity and Ecosystem Services

REMEMBER

LESSON 1.1
Week 1 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.1: Variables, Data, and Scientific Explanations

TODAY'S GOAL

I can identify independent, dependent, and controlled variables and write a scientific explanation using CER.

Standards: SEP 3, 4, 6

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.1 in this book with your teacher.
- [ ] Do the Investigate activity: Temperature and Dissolving.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.1, Part A.
- [ ] Complete the rest of Workbook Lesson 1.1 (Parts B–E).

1 Get Ready

I WONDER

Does a fizzing tablet dissolve faster in hot water or cold water? How could you find out fairly?

2 Learn

A controlled experiment tests how one factor affects another. The independent variable is what you change. The dependent variable is what you measure. Everything else

WORDS & MORE

WORDS TO KNOW

independent variable
The variable the scientist changes on purpose.

dependent variable
The variable that is measured; it responds to the change.

controlled variables
Conditions kept the same so the test is fair.

hypothesis
A testable prediction that explains how one variable affects another.

REMEMBER

stays the same as controlled variables. A good hypothesis is written so it can be tested: “If..., then..., because....”

Part	Fizzing-tablet experiment
question	How does water temperature affect how long a tablet takes to dissolve?
hypothesis	If the water is warmer, then the tablet will dissolve faster, because particles move faster at higher temperatures.
independent variable	water temperature
dependent variable	time to dissolve (seconds)
controlled variables	amount of water, type and size of tablet, cup, no stirring

WORDS & MORE

INVESTIGATE · TEMPERATURE AND DISSOLVING

How does water temperature affect dissolving time?

You need: 3 identical effervescent antacid tablets (or 3 equal spoonfuls of sugar); 3 clear cups; cold, room-temperature, and warm tap water; thermometer; stopwatch.

1. Measure 200 mL of water into each cup and record each temperature.
2. Drop one tablet into each cup at the same moment and start the timer.
3. Record the time when each tablet is completely dissolved.
4. Repeat if you can, and average the trials.

Record: a data table and a line or bar graph with temperature on the x-axis and time on the y-axis.

Safety: Use warm tap water, not boiling water. Do not swallow the tablets or solutions.

KEY IDEA

Change one independent variable, measure the dependent variable, control everything else, and explain results with CER.

EXPLAIN IT · WRITE A SCIENTIFIC EXPLANATION

Data: 8°C → 95 s; 21°C → 52 s; 40°C → 24 s

1. Claim: Tablets dissolve faster in warmer water.
2. Evidence: dissolving time dropped from 95 s at 8°C to 52 s at 21°C to 24 s at 40°C.
3. Reasoning: particles in warmer water move faster, so they collide with the tablet more often and break it apart sooner.

REMEMBER

✓ REMEMBER

Independent is changed, dependent is measured, controlled stay the same; explain with CER.

Explanation: A complete explanation connects the evidence to a scientific idea.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.1, Part A. Then check with your teacher.

1. Plant height is measured after different amounts of fertilizer. Independent variable?
2. Same experiment: dependent variable?
3. Name two controlled variables.

WORDS & MORE

REMEMBER

LESSON 1.2
Week 1 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.2: Earth's Spheres and the Water Cycle

TODAY'S GOAL

I can describe Earth's four spheres and model how energy from the sun and gravity drive the water cycle.

Standards: MS-ESS2-4

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.2 in this book with your teacher.
- [ ] Do the Investigate activity: Water Cycle in a Bag.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.2, Part A.
- [ ] Complete the rest of Workbook Lesson 1.2 (Parts B–D).

1 Get Ready

I WONDER

The water you drank today may once have been in a dinosaur, a cloud, or an ocean. How could that be?

2 Learn

Scientists study Earth as a system of four interacting parts, or spheres: the geosphere, hydrosphere, atmosphere, and biosphere. Matter, like water, constantly moves among them. Earth has about the same amount of water today as it had long ago; it just keeps cycling.

WORDS & MORE

WORDS TO KNOW

geosphere
The solid rock and soil of Earth.

hydrosphere
All of Earth's water.

atmosphere
The layer of gases around Earth.

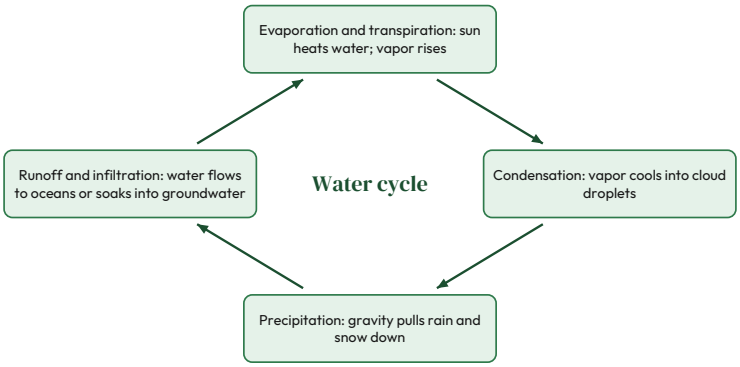
biosphere
All living things on Earth.

transpiration
The release of water vapor from plant leaves.

condensation
The change from water vapor to liquid water.

REMEMBER

WORDS & MORE



Two forces drive the cycle. Energy from the sun evaporates water and powers winds. Gravity pulls precipitation down and makes water flow downhill into rivers, lakes, and oceans, and soak into the ground.

KEY IDEA

Water cycles among Earth's spheres, driven by energy from the sun and by gravity.

INVESTIGATE · WATER CYCLE IN A BAG

Can you model the water cycle?

You need: a zip-top bag; 1/4 cup water with a few drops of blue food coloring; tape; a sunny window; marker.

1. Draw a sun, cloud, and waves on the bag.
2. Pour in the water and seal the bag.
3. Tape it to a sunny window.
4. Observe it after 1 hour, after 1 day, and after 2 days.

Record: observations and a labeled sketch showing evaporation, condensation, and precipitation in the bag.

✓ REMEMBER

Water cycles among Earth's spheres, powered by the sun and gravity.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.2, Part A. Then check with your teacher.

1. Which sphere includes clouds and air?
2. What process returns water vapor from plants?
3. What force makes rain fall and rivers flow?

REMEMBER

LESSON 1.3
Week 2 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.3: The Carbon and Oxygen Cycles

TODAY'S GOAL

I can explain how carbon and oxygen cycle among living things, the atmosphere, the oceans, and rocks.

Standards: MS-LS2-3

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.3 in this book with your teacher.
- [ ] Do the Investigate activity: Follow a Carbon Atom.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.3, Part A.
- [ ] Complete the rest of Workbook Lesson 1.3 (Parts B–D).

1 Get Ready

I WONDER

Carbon is in your body, in a diamond, in soda bubbles, and in the air. How can one element be in all of these?

2 Learn

Carbon is the building block of living things. Carbon atoms are never used up; they cycle. Plants take carbon dioxide from the air during photosynthesis and build sugars. Animals eat plants. Living things release carbon dioxide during cellular respiration, and decomposers release it as

WORDS & MORE

WORDS TO KNOW

carbon cycle
The movement of carbon among the air, living things, oceans, and rocks.

decomposer
An organism that breaks down dead matter and returns materials to the environment.

fossil fuel
Coal, oil, or natural gas formed from ancient living things.

combustion
Burning; a reaction with oxygen that releases energy and carbon dioxide.

REMEMBER

they break down dead matter. Oceans absorb and release carbon dioxide too. Over millions of years, some carbon becomes buried as fossil fuels; burning them releases that carbon quickly.

The oxygen cycle is linked to the carbon cycle: photosynthesis releases oxygen, and respiration and combustion use oxygen. Today air is about 21 percent oxygen and about 0.04 percent carbon dioxide.

Process	Carbon dioxide	Oxygen
photosynthesis	taken in	released
cellular respiration	released	taken in
decomposition	released	taken in
combustion (burning fuels)	released	taken in
ocean absorption	taken in (dissolves in water)	—

WORDS & MORE

KEY IDEA

Photosynthesis removes carbon dioxide and adds oxygen; respiration, decomposition, and combustion do the opposite.

INVESTIGATE · FOLLOW A CARBON ATOM

What paths can one carbon atom take?

You need: station cards: Atmosphere, Plant, Animal, Decomposer, Ocean, Fossil Fuel; one die.

1. Start at the Atmosphere station.
2. Roll the die; each station card tells where a roll sends you and why (e.g. 1–3: a plant takes you in during photosynthesis).
3. Record each move for 10 rolls.
4. Compare paths with a partner.

Record: your atom's path as a flow chart, with the process that moved it at each arrow.

✓ REMEMBER

Carbon and oxygen cycle through photosynthesis, respiration, decomposition, combustion, and oceans.

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.3, Part A. Then check with your teacher.

1. Which process removes carbon dioxide from the air?
2. Name two processes that release carbon dioxide.
3. About what percent of air is oxygen?

REMEMBER

LESSON 1.4
Week 2 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.4: The Nitrogen Cycle

TODAY'S GOAL

I can explain how nitrogen moves from the air into living things and back.

Standards: MS-LS2-3

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.4 in this book with your teacher.
- [ ] Do the Investigate activity: Root Nodule Search.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.4, Part A.
- [ ] Complete the rest of Workbook Lesson 1.4 (Parts B–D).

1 Get Ready

I WONDER

Air is about 78 percent nitrogen, and your body needs nitrogen to build proteins. Why can't you just breathe it in and use it?

2 Learn

Living things need nitrogen to build proteins and DNA, but most organisms can't use nitrogen gas (N_2) directly. It has to be fixed into compounds such as ammonia and nitrates.

WORDS & MORE

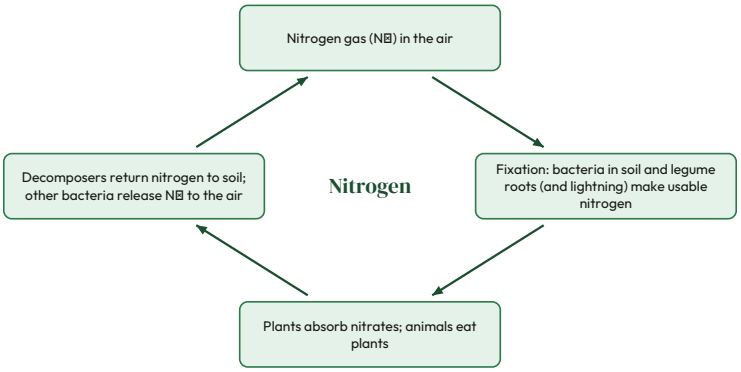
WORDS TO KNOW

nitrogen fixation
Changing nitrogen gas into a form plants can use.

legume
A plant such as a bean, pea, or peanut that hosts nitrogen-fixing bacteria in its roots.

REMEMBER

WORDS & MORE



Farmers use this cycle. Planting legumes such as soybeans or peanuts adds usable nitrogen to the soil. Peanut farmer and scientist George Washington Carver taught Southern farmers to rotate cotton with peanuts and other legumes to restore worn-out soil.

KEY IDEA

Bacteria fix nitrogen gas into usable forms; plants absorb it, animals eat plants, and decomposers and bacteria return it.

INVESTIGATE · ROOT NODULE SEARCH

Where do nitrogen-fixing bacteria live?

You need: a bean or pea plant (or photos from your teacher); magnifying glass; water.

1 Gently dig up the plant and rinse the soil from its roots.

2 Look for small round bumps (nodules) on the roots.

3 Sketch the roots and nodules.

Record: a labeled sketch and an explanation of what the bacteria in the nodules do.

✓ REMEMBER

Bacteria fix nitrogen so plants and animals can use it; decomposers return it.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.4, Part A. Then check with your teacher.

1. What percent of air is nitrogen?
2. What does nitrogen fixation do?
3. Why do farmers plant legumes?

REMEMBER

LESSON 1.5
Week 3 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.5: Photosynthesis

TODAY'S GOAL

I can explain how plants use light energy to make sugar and oxygen from carbon dioxide and water.

Standards: MS-LS1-6

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.5 in this book with your teacher.
- [ ] Do the Investigate activity: Floating Leaf Disks.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.5, Part A.
- [ ] Complete the rest of Workbook Lesson 1.5 (Parts B–D).

1 Get Ready

I WONDER

A huge oak tree grows from a tiny acorn. Where does all that mass come from?

2 Learn

Surprisingly, most of a tree's mass comes from the air! In photosynthesis, chloroplasts in leaf cells use light energy captured by chlorophyll to combine carbon dioxide from the air with water from the soil. They produce glucose, which the plant uses for energy and to build its body, and release oxygen.

WORDS & MORE

WORDS TO KNOW

photosynthesis
The process plants use to turn light energy, carbon dioxide, and water into sugar and oxygen.

chloroplast
The part of a plant cell where photosynthesis happens.

chlorophyll
The green pigment that captures light energy.

glucose
A sugar made in photosynthesis that stores energy.

REMEMBER

WORDS & MORE

Word equation	carbon dioxide + water —(light energy)→ glucose + oxygen
Chemical equation	$6CO_2 + 6H_2O \xrightarrow{light} C_6H_{12}O_6 + 6O_2$

Photosynthesis changes light energy into chemical energy stored in sugar. Almost every food chain on Earth begins with this process.

KEY IDEA

Photosynthesis: carbon dioxide + water + light energy → glucose + oxygen, in chloroplasts.

INVESTIGATE · FLOATING LEAF DISKS

Does light intensity affect the rate of photosynthesis?

You need: fresh spinach leaves; a straw or hole punch; a plastic syringe (no needle); baking soda solution (1/8 tsp in 300 mL water) with 1 drop dish soap; 2 clear cups; a lamp.

1. Punch 20 disks from the leaves.
2. Put 10 disks in the syringe with solution, push out the air, cover the tip, and pull the plunger to remove gas from the leaves until the disks sink. Repeat for the other 10.
3. Put 10 sunk disks in a cup under the lamp and 10 in a cup in a dark place.
4. Count floating disks every minute for 15 minutes.

Record: a line graph of floating disks vs time for both cups, and a CER explaining the difference (oxygen produced in photosynthesis makes disks float).

✓ REMEMBER

Plants turn carbon dioxide and water into glucose and oxygen using light.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.5, Part A. Then check with your teacher.

1. Where does most of a tree's mass come from?
2. What are the products of photosynthesis?
3. What pigment captures light?

REMEMBER

LESSON 1.6
Week 3 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.6: Cellular Respiration

TODAY'S GOAL

I can explain how cells release energy from food through cellular respiration.

Standards: MS-LS1-7

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.6 in this book with your teacher.
- [ ] Do the Investigate activity: Yeast Balloons.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.6, Part A.
- [ ] Complete the rest of Workbook Lesson 1.6 (Parts B–D).

1 Get Ready

I WONDER

You ate breakfast hours ago. How is that food powering your brain right now?

2 Learn

All organisms, including plants, need energy. In cellular respiration, mitochondria in cells break down glucose with oxygen, releasing energy the cell can use. Carbon dioxide and water are given off as waste.

WORDS & MORE

WORDS TO KNOW

cellular respiration
The process cells use to release energy from glucose using oxygen.

mitochondria
Cell parts where cellular respiration releases energy.

fermentation
A process that releases energy from sugar without oxygen.

REMEMBER

Word equation	glucose + oxygen → carbon dioxide + water + energy
Chemical equation	$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy

Notice that respiration is roughly the reverse of photosynthesis. The products of one are the reactants of the other, so matter cycles between plants and animals while energy flows from the sun, through living things, and is eventually lost as heat.

Some organisms, like yeast, can also release energy without oxygen through fermentation, giving off carbon dioxide. That's why bread dough rises.

WORDS & MORE

KEY IDEA

Respiration: glucose + oxygen → carbon dioxide + water + energy, in mitochondria. It is the reverse of photosynthesis.

INVESTIGATE · YEAST BALLOONS

Does yeast need sugar to release carbon dioxide?

You need: 2 empty water bottles; 2 balloons; 2 packets of dry yeast; warm water; 2 tablespoons sugar.

1. Add 1 cup warm water and 1 packet yeast to each bottle.
2. Add sugar to bottle A only. Swirl both.
3. Stretch a balloon over each bottle's mouth.
4. Measure each balloon's circumference every 10 minutes for 30 minutes.

Record: a data table and a graph; explain what filled the balloon and why one grew more.

✓ REMEMBER

Cells release energy from glucose with oxygen; respiration and photosynthesis cycle matter.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.6, Part A. Then check with your teacher.

1. Where does cellular respiration happen?
2. What gas filled the balloon?
3. Do plants carry out respiration?

REMEMBER

LESSON 1.7
Week 4 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.7: Food Webs and Energy Pyramids

TODAY'S GOAL

I can model how energy flows through a food web and explain why less energy reaches each higher level.

Standards: MS-LS2-3

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.7 in this book with your teacher.
- [ ] Do the Investigate activity: Build a Food Web.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.7, Part A.
- [ ] Complete the rest of Workbook Lesson 1.7 (Parts B–D).

1 Get Ready

I WONDER

Why are there far more grasshoppers than hawks in a field?

2 Learn

Energy enters most ecosystems as sunlight captured by producers. Primary consumers (herbivores) eat producers, secondary consumers eat herbivores, and so on. Decomposers break down dead organisms at every level. Arrows in a food chain point in the direction energy flows:

WORDS & MORE

WORDS TO KNOW

producer
An organism that makes its own food, usually by photosynthesis.

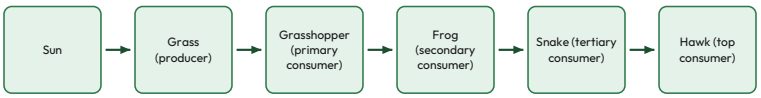
consumer
An organism that eats other organisms.

food web
A diagram of all the feeding relationships in an ecosystem.

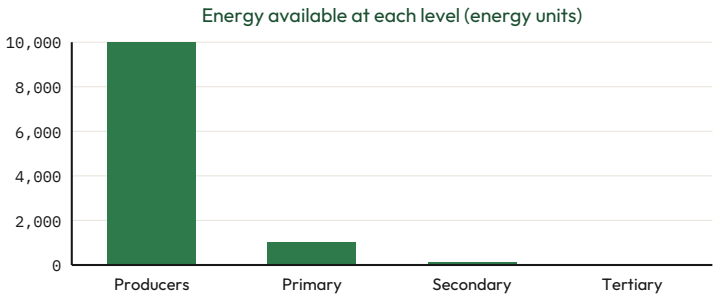
trophic level
A feeding level in a food chain or energy pyramid.

REMEMBER

from the eaten to the eater.



At each level, organisms use most of their energy to live, move, and stay warm, and much is lost as heat. On average only about 10 percent of the energy at one level is stored in the next. That's why an energy pyramid narrows at the top and why there are fewer top predators.



KEY IDEA

Energy flows from producers to consumers; only about 10 percent passes to each higher level, while matter is recycled by decomposers.

INVESTIGATE · BUILD A FOOD WEB

How are organisms in a forest ecosystem connected?

You need: cards: oak tree, grass, acorn weevil, grasshopper, mouse, rabbit, deer, frog, snake, hawk, fox, mushroom (decomposer); yarn or a pencil.

1 Arrange the cards on a large sheet of paper.

2 Draw arrows from each food to the organism that eats it.

3 Label producers, consumers, and the decomposer.

4 Remove the mouse. Predict two effects on the web.

Record: your food web and your predictions.

✓ REMEMBER

Energy flows up through feeding levels with about 10 percent passed on; matter recycles.

WORDS & MORE

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.7, Part A. Then check with your teacher.

1. Which way do food-chain arrows point?
2. If grass has 20,000 units, about how much reaches grasshoppers?
3. Why are top predators few?

REMEMBER

LESSON 1.8
Week 4 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.8: Biodiversity and Ecosystem Services

TODAY'S GOAL

I can explain why biodiversity matters and evaluate ways to protect ecosystem services.

Standards: MS-LS2-5

TODAY'S ASSIGNMENT

- [ ] Read Lesson 1.8 in this book with your teacher.
- [ ] Do the Investigate activity: Biodiversity Survey.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.8, Part A.
- [ ] Complete the rest of Workbook Lesson 1.8 (Parts B–D).

1 Get Ready

I WONDER

What would happen to your breakfast if all the bees disappeared?

2 Learn

Biodiversity is the variety of species, genes, and ecosystems in a place. Ecosystems with high biodiversity are usually more stable: if one species declines, others can fill its role.

WORDS & MORE

WORDS TO KNOW

biodiversity
The variety of living things in an area.

ecosystem services
Benefits people get from healthy ecosystems, such as clean water and pollination.

pollinator
An animal, such as a bee, that carries pollen between flowers.

REMEMBER

Ecosystem service	Example
pollination	Bees, butterflies, and bats pollinate many fruits, nuts, and vegetables.
clean water	Wetlands filter pollutants as water flows through them.
flood control	Wetlands and forests soak up heavy rain like sponges.
healthy soil	Decomposers and earthworms recycle nutrients.
food and materials	Fish, timber, and medicines come from ecosystems.

WORDS & MORE

EXPLAIN IT · EVALUATE A DESIGN SOLUTION

A town wants to reduce flooding. Option A: build a concrete channel. Option B: restore a wetland upstream.

1. Criteria: reduce flooding, cost, protect biodiversity, clean water.
2. A: moves water fast but removes habitat and doesn't filter water.
3. B: soaks up and slows water, filters pollutants, and provides habitat, but needs land.

Explanation: Option B meets more criteria, especially biodiversity and clean water, though land and cost are constraints.

KEY IDEA

Biodiversity keeps ecosystems stable and provides services people depend on; good design solutions protect it.

INVESTIGATE · BIODIVERSITY SURVEY

How many kinds of living things live in a small area?

You need: 4 sticks and string to mark a 1 m × 1 m square; magnifying glass; paper and a pencil.

1. Mark a square in a lawn and another in a less-mowed area (garden edge, field, or woods).
2. Count the different kinds of plants and animals in each square for 10 minutes.
3. Compare the two squares.

Record: a table of species counts and an explanation of which area has more biodiversity and why.

✓ REMEMBER

Biodiversity stabilizes ecosystems and provides services worth protecting.

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 1.8, Part A. Then check with your teacher.

1. What is biodiversity?
2. Name two ecosystem services.
3. Why is a diverse ecosystem more stable?

UNIT 1 REVIEW

Unit 1 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember...
1.1	Independent is changed, dependent is measured, controlled stay the same; explain with CER.
1.2	Water cycles among Earth's spheres, powered by the sun and gravity.
1.3	Carbon and oxygen cycle through photosynthesis, respiration, decomposition, combustion, and oceans.
1.4	Bacteria fix nitrogen so plants and animals can use it; decomposers return it.
1.5	Plants turn carbon dioxide and water into glucose and oxygen using light.
1.6	Cells release energy from glucose with oxygen; respiration and photosynthesis cycle matter.
1.7	Energy flows up through feeding levels with about 10 percent passed on; matter recycles.
1.8	Biodiversity stabilizes ecosystems and provides services worth protecting.

UNIT 2

Properties and Changes of Matter

Atoms and molecules, physical properties and states, changes of state, density, solubility and conductivity, chemical properties, and chemical changes.

Everything around you, from the air you breathe to the phone in your hand, is made of tiny atoms. In this unit you'll build models of molecules, identify substances by their properties, explain how particles behave when matter melts or boils, calculate density, separate mixtures, and finally test whether matter can ever disappear in a chemical reaction.

In this unit you will…

- Model atoms, molecules, and compounds.
- Describe and measure physical properties, including density.
- Explain changes of state in terms of particle motion and energy.
- Distinguish physical and chemical properties and changes using evidence.
- Test conservation of mass in a chemical reaction.

SCIENCE IN YOUR WORLD

Chemists, chefs, pharmacists, and materials engineers use the properties of matter every day to make medicines, cook food, design safer materials, and recycle.

UNIT VOCABULARY

atom	The smallest particle of an element.
compound	Different elements chemically joined.
density	Mass per volume.
thermal energy	The energy of moving particles.
chemical change	A change that makes new substances.
conservation of mass	Mass is not created or destroyed in a reaction.

Lessons in this unit

Lesson	Title
2.1	Atoms, Elements, and Molecules
2.2	Physical Properties and States of Matter
2.3	Particles in Motion: Changes of State
2.4	Density
2.5	Solubility, Conductivity, and Mixtures
2.6	Chemical Properties
2.7	Physical and Chemical Changes
2.8	Lab: Evidence of a Chemical Reaction

REMEMBER

LESSON 2.1
Week 1 · Lesson 1 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.1: Atoms, Elements, and Molecules

TODAY'S GOAL

I can describe atoms, elements, molecules, and compounds and model simple molecules.

Standards: MS-PS1-1

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.1 in this book with your teacher.
- [ ] Do the Investigate activity: Model Molecules.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.1, Part A.
- [ ] Complete the rest of Workbook Lesson 2.1 (Parts B–E).

1 Get Ready

I WONDER

If you cut a piece of aluminum foil in half, then in half again, over and over, would you ever reach a piece you couldn't cut?

2 Learn

All matter, from air to rocks to you, is made of tiny particles called atoms. Atoms are so small that a single grain of sand contains more than a billion billion of them. An element is a pure substance made of only one kind of atom. Scientists

WORDS & MORE

WORDS TO KNOW

matter
Anything that has mass and takes up space.

atom
The smallest particle of an element that still has the properties of that element.

element
A pure substance made of only one kind of atom.

molecule
Two or more atoms joined together.

compound
A substance made of two or more different elements chemically joined in a set ratio.

REMEMBER

know of 118 elements, organized on the periodic table. Each has a one- or two-letter symbol, such as O for oxygen, C for carbon, and Fe for iron.

Atoms can join to form molecules. An oxygen molecule (O_2) is two oxygen atoms. When atoms of different elements join in a fixed ratio, they form a compound. Water (H_2O) is always two hydrogen atoms joined to one oxygen atom. A compound has properties very different from its elements: hydrogen and oxygen are gases, but together they make liquid water.

Substance	Formula	Atoms in one molecule	Element or compound?
oxygen gas	O_2	2 oxygen	element
water	H_2O	2 hydrogen, 1 oxygen	compound
carbon dioxide	CO_2	1 carbon, 2 oxygen	compound
table sugar (sucrose)	$C_{12}H_{22}O_{11}$	12 carbon, 22 hydrogen, 11 oxygen	compound

WORDS & MORE

EXPLAIN IT · READ A CHEMICAL FORMULA

How many atoms are in one molecule of ammonia, NH_3?

1 N stands for nitrogen. With no small number after it, there is 1 nitrogen atom.

2 H stands for hydrogen. The subscript 3 means 3 hydrogen atoms.

3 Add them: 1 + 3 = 4.

Explanation: One molecule of NH_3 has 4 atoms: 1 nitrogen and 3 hydrogen.

KEY IDEA

All matter is made of atoms. Elements contain one kind of atom; molecules are atoms joined together; compounds join different elements in a fixed ratio and have new properties.

INVESTIGATE · MODEL MOLECULES

How can models show the structure of molecules?

You need: mini marshmallows and gumdrops (or clay in two or three colors); toothpicks.

1 Choose a color for each element: for example, white = hydrogen, red = oxygen, black = carbon.

2 Build H_2O, O_2, and CO_2 using toothpicks as bonds.

3 Build CH_4 (methane): one carbon in the center with four hydrogens.

4 Sketch each model and write its formula.

REMEMBER

WATCH OUT

The subscript belongs to the atom just before it. In CO_2, the 2 means two oxygen atoms, not two carbon atoms.

✓ REMEMBER

Matter is made of atoms; molecules and compounds are atoms joined together.

Record: labeled sketches of four molecule models with a color key.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.1, Part A. Then check with your teacher.

1. What is the difference between an element and a compound?
2. How many atoms are in one molecule of CO_2?
3. Is O_2 an element or a compound?

WORDS & MORE

REMEMBER

LESSON 2.2
Week 1 · Lesson 2 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.2: Physical Properties and States of Matter

TODAY'S GOAL

I can describe physical properties and compare solids, liquids, and gases.

Standards: MS-PS1-2, MS-PS1-4

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.2 in this book with your teacher.
- [ ] Do the Investigate activity: Property Detective.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.2, Part A.
- [ ] Complete the rest of Workbook Lesson 2.2 (Parts B–E).

1 Get Ready

I WONDER

How could you tell the difference between salt and sugar without tasting them?

2 Learn

A physical property can be observed or measured without changing the substance into something new. Color, texture, hardness, odor, mass, volume, melting point, boiling point, and whether a metal is malleable are all physical properties. Scientists use them to identify substances.

WORDS & MORE

WORDS TO KNOW

physical property
A property you can observe or measure without changing what the substance is.

state of matter
The form matter takes: solid, liquid, or gas.

volume
The amount of space something takes up.

mass
The amount of matter in an object.

malleable
Able to be hammered or pressed into shapes without breaking.

REMEMBER

The most familiar physical property is state. Matter on Earth is usually a solid, liquid, or gas. The difference is in how the particles are arranged and how they move.

	Solid	Liquid	Gas
shape	definite	takes the shape of its container	fills its container
volume	definite	definite	changes to fill the container
particles	packed tightly; vibrate in place	close together; slide past each other	far apart; move fast in all directions
example	ice	water	water vapor (steam)

WORDS & MORE

EXPLAIN IT · IDENTIFY SUBSTANCES BY PROPERTIES

Two white powders: A dissolves easily and its crystals are cubes; B melts and turns brown when heated in a pan.

1 Cube-shaped crystals are a property of table salt.

2 Melting and browning when heated is a property of sugar.

3 Properties, not appearance alone, identify each one.

Explanation: A is salt; B is sugar.

KEY IDEA

Physical properties can be observed without changing a substance. Solids, liquids, and gases differ in shape, volume, and how their particles move.

INVESTIGATE · PROPERTY DETECTIVE

Which properties can identify an unknown substance?

You need: small samples of salt, sugar, baking soda, and cornstarch; a magnifying glass; cups of water; vinegar; a data table.

1 Observe each powder with the magnifying glass and describe its crystals.

2 Rub a pinch between your fingers and describe its texture.

3 Stir a spoonful into water: does it dissolve?

4 Have someone give you a mystery sample and identify it from your table.

Record: a properties table and the identity of the mystery sample, with your evidence.

Safety: Don't taste any samples. Wear safety goggles.

✓ REMEMBER

Physical properties identify substances; states differ in particle arrangement and motion.

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.2, Part A. Then check with your teacher.

1. Name three physical properties.
2. Which state has a definite volume but no definite shape?
3. How do gas particles move?

REMEMBER

LESSON 2.3
Week 2 · Lesson 1 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.3: Particles in Motion: Changes of State

TODAY'S GOAL

I can explain how adding or removing thermal energy changes particle motion, temperature, and state.

Standards: MS-PS1-4

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.3 in this book with your teacher.
- [ ] Do the Investigate activity: Ice Melt Data.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.3, Part A.
- [ ] Complete the rest of Workbook Lesson 2.3 (Parts B–E).

1 Get Ready

I WONDER

Why does a puddle disappear on a sunny day, even though it never boils?

2 Learn

Particles of matter are always moving. Adding thermal energy makes them move faster; removing it slows them down. Temperature measures the average motion energy

WORDS & MORE

WORDS TO KNOW

thermal energy
The total energy of the moving particles in a substance.

melting point
The temperature at which a solid becomes a liquid.

evaporation
The change from liquid to gas at the surface of a liquid.

condensation
The change from gas to liquid.

sublimation
The change from solid directly to gas.

REMEMBER

of the particles.

When enough energy is added, particles move fast enough to break free of their arrangement, and the state changes. While a substance is changing state, its temperature stays the same: the added energy goes into separating the particles, not speeding them up. That's why a glass of ice water stays near 0°C until all the ice melts.

Change	From → to	Energy	Example
melting	solid → liquid	added	an ice pop dripping
freezing	liquid → solid	removed	water becoming ice cubes
evaporation / boiling	liquid → gas	added	a puddle drying; a pot boiling
condensation	gas → liquid	removed	drops on a cold glass
sublimation	solid → gas	added	dry ice making fog

WORDS & MORE

EXPLAIN IT · READ A HEATING CURVE

Ice at −10°C is heated steadily. The temperature rises to 0°C, stays flat for 4 minutes, then rises again to 100°C and stays flat.

1 Rising sections: particles speed up and temperature increases.

2 First flat section at 0°C: melting; energy separates particles.

3 Second flat section at 100°C: boiling.

Explanation: Flat parts of a heating curve show changes of state.

KEY IDEA

Adding thermal energy speeds particles up; removing it slows them down. During a change of state, temperature stays constant while particles rearrange.

INVESTIGATE · ICE MELT DATA

What happens to temperature while ice melts?

You need: a cup of crushed ice; a thermometer; a timer; a stirring spoon.

1 Measure the starting temperature of the ice.

2 Let it sit at room temperature; stir and record the temperature every 2 minutes.

3 Note the time when the ice is half melted and fully melted.

4 Keep recording for 10 more minutes.

REMEMBER

✓ REMEMBER

Thermal energy changes particle motion and can change state; temperature holds steady during the change.

Record: a data table and a line graph of temperature over time, with the melting section labeled.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.3, Part A. Then check with your teacher.

1. What happens to particle motion when thermal energy is added?
2. Name the change from gas to liquid.
3. Why does the temperature stay at 0°C while ice melts?

WORDS & MORE

REMEMBER

LESSON 2.4
Week 2 · Lesson 2 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.4: Density

TODAY'S GOAL

I can calculate density and use it to predict whether an object will sink or float.

Standards: MS-PS1-2

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.4 in this book with your teacher.
- [ ] Do the Investigate activity: Density Column.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.4, Part A.
- [ ] Complete the rest of Workbook Lesson 2.4 (Parts B–E).

1 Get Ready

I WONDER

A huge steel ship floats, but a small steel nail sinks. How can that be?

2 Learn

Density tells how much mass is packed into a certain volume. It's a physical property, and every pure substance has its own density, so density can help identify a material.

Density = mass ÷ volume, or D = m ÷ V. Mass is measured in grams (g) and volume in cubic centimeters (cm^3) or milliliters (mL), so density is in g/cm^3 or g/mL. To find the volume of an irregular object, use displacement: measure how much the water level rises when you drop it in.

WORDS & MORE

WORDS TO KNOW

density
How much mass is packed into a given volume.

displacement
Measuring volume by how much water an object pushes aside.

buoyancy
The upward push of a liquid on an object in it.

REMEMBER

Substance	Density (g/cm³)	In water, it…
cork	about 0.24	floats
ice	0.92	floats
vegetable oil	about 0.92	floats on water
water	1.00	—
aluminum	2.70	sinks
iron	7.87	sinks
gold	19.3	sinks

An object less dense than water floats; one more dense sinks. A steel ship floats because its hollow shape is full of air, so the ship's overall density is less than water's.

EXPLAIN IT · CALCULATE DENSITY

A metal block has a mass of 54 g and a volume of 20 cm³. What is it made of?

1. Write the formula: D = m ÷ V.
2. Substitute: D = 54 g ÷ 20 cm³.
3. Divide: D = 2.7 g/cm³.
4. Compare with the table: 2.7 g/cm³ matches aluminum.

Explanation: The block is aluminum, with a density of 2.7 g/cm³.

INVESTIGATE · DENSITY COLUMN

Can liquids of different densities stack?

You need: a tall clear glass; honey, dish soap, water with food coloring, vegetable oil; small objects: a grape, a cork, a plastic bead.

1. Slowly pour about 2 cm of each liquid, in this order: honey, dish soap, colored water, oil.
2. Watch whether the layers mix.
3. Gently drop in each small object.
4. Record where each object stops.

Record: a labeled drawing of your column from densest (bottom) to least dense (top), with each object's position.

KEY IDEA

Density = mass ÷ volume. Objects less dense than water float, and denser objects sink. Density helps identify substances.

WATCH OUT

Density is not the same as weight. A big log weighs more than a small pebble, but the log floats because wood is less dense than water.

WORDS & MORE

REMEMBER

✓ REMEMBER

Density = mass ÷ volume; it predicts sinking and floating and identifies substances.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.4, Part A. Then check with your teacher.

1. A rock has a mass of 30 g and a volume of 10 cm^3. What is its density?
2. Will an object with a density of 0.6 g/cm^3 float in water?
3. How can you find the volume of an irregular object?

WORDS & MORE

REMEMBER

LESSON 2.5
Week 3 · Lesson 1 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.5: Solubility, Conductivity, and Mixtures

TODAY'S GOAL

I can describe solubility and conductivity and tell mixtures and solutions apart.

Standards: MS-PS1-2

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.5 in this book with your teacher.
- [ ] Do the Investigate activity: Separate a Mixture.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.5, Part A.
- [ ] Complete the rest of Workbook Lesson 2.5 (Parts B–E).

1 Get Ready

I WONDER

Why can you stir sugar into tea until it disappears, but sand just sinks to the bottom?

2 Learn

A mixture contains two or more substances that are not chemically joined. Each keeps its own properties, and you can separate them by physical means, such as filtering, using a magnet, or evaporating. Trail mix and sand in water are mixtures you can see. A solution is a mixture so evenly

WORDS & MORE

WORDS TO KNOW

solubility
How much of a substance can dissolve in another.

solution
A mixture in which one substance is spread evenly through another.

mixture
Two or more substances mixed together but not chemically joined.

conductivity
How well a material lets heat or electricity pass through it.

solute
The substance that dissolves in a solution.

REMEMBER

blended that it looks like one substance, such as salt water or air.

Solubility describes how much of a substance, the solute, can dissolve in another, the solvent. Sugar and salt are very soluble in water; sand is not. Solubility usually increases with temperature: you can dissolve more sugar in hot tea than in iced tea.

Conductivity describes how well a material lets heat or electricity pass through. Metals such as copper are excellent conductors, which is why electric wires are made of copper. Rubber, plastic, and glass are insulators. Pure water conducts electricity poorly, but salt water conducts it well.

Mixture	Type	How to separate
sand and water	mixture (not a solution)	filter
iron filings and sand	mixture	magnet
salt water	solution	evaporate the water
air	solution of gases	cool it until gases condense (in industry)

KEY IDEA

Mixtures keep their parts' properties and can be separated physically. Solubility and conductivity are physical properties that help identify and separate substances.

INVESTIGATE · SEPARATE A MIXTURE

Can you separate sand, salt, and iron filings using their properties?

You need: a mixture of sand, salt, and iron filings (or paper clips); a magnet in a plastic bag; a coffee filter and funnel; water; a shallow dish.

1. Pass the magnet over the mixture to remove the iron.
2. Stir the rest into warm water so the salt dissolves.
3. Pour it through the filter to catch the sand.
4. Leave the salty water in the dish for a few days to evaporate.

Record: a flowchart of your steps, naming the property used at each step.

Safety: Wear goggles. Keep the iron filings in a bag on the magnet.

✓ REMEMBER

Mixtures can be separated by properties such as solubility and magnetism; conductivity is another key property.

WORDS & MORE

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.5, Part A. Then check with your teacher.

1. Is salt water a mixture or a compound?
2. How could you separate iron filings from sand?
3. Does sugar usually dissolve better in hot or cold water?

REMEMBER

LESSON 2.6
Week 3 · Lesson 2 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.6: Chemical Properties

TODAY'S GOAL

I can describe chemical properties and explain how they differ from physical properties.

Standards: MS-PS1-2

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.6 in this book with your teacher.
- [ ] Do the Investigate activity: Rust Race.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.6, Part A.
- [ ] Complete the rest of Workbook Lesson 2.6 (Parts B–E).

1 Get Ready

I WONDER

A wooden log and a metal pipe are both solid and brown. Which one would you put in a fireplace? How do you know?

2 Learn

A chemical property describes how a substance can change into a new substance. Unlike physical properties, you can only observe a chemical property by changing the substance. Once wood burns, it's no longer wood: it has become ash, smoke, water vapor, and carbon dioxide.

WORDS & MORE

WORDS TO KNOW

chemical property
A property that describes how a substance can change into a new substance.

flammability
How easily a substance burns.

reactivity
How easily a substance reacts with other substances.

oxidation
A reaction with oxygen, such as rusting or burning.

REMEMBER

Chemical property	What it describes	Example
flammability	how easily it burns	paper burns easily; water doesn't
reactivity with oxygen	whether it rusts or tarnishes	iron rusts; gold doesn't
reactivity with acids	whether it fizzes or dissolves in acid	baking soda fizzes in vinegar
reactivity with water	whether it reacts in water	sodium metal reacts violently; copper doesn't

Chemical properties help people choose materials safely. Gold and silver are used in jewelry and electronics partly because they don't react easily with oxygen. Iron is strong but rusts, so steel bridges are painted or coated. Flammable liquids, such as gasoline, must be stored away from flames.

WORDS & MORE

EXPLAIN IT · PHYSICAL OR CHEMICAL PROPERTY?

Decide for each: (a) copper is shiny; (b) copper turns green outdoors over time; (c) copper melts at 1,085°C.

1 (a) Shininess can be observed without changing copper: physical.

2 (b) Turning green means copper reacted with air and water to form a new substance: chemical.

3 (c) Melted copper is still copper: physical.

Explanation: (a) physical, (b) chemical, (c) physical.

KEY IDEA

Chemical properties, such as flammability and reactivity, describe how a substance can change into a new substance. You observe them only by changing the substance.

INVESTIGATE · RUST RACE

Which conditions make iron rust fastest?

You need: four steel wool pads or iron nails; four cups; water, salt water, vinegar, and an empty (dry) cup; labels.

1 Place one pad or nail in each cup: dry, water, salt water, vinegar.

2 Label the cups and predict which will rust fastest.

3 Observe each day for five days.

4 Record color, texture, and amount of rust.

Record: a daily observation table and a claim about which condition caused the most rust.

Safety: Wear goggles and gloves. Wash hands after handling steel wool.

✓ REMEMBER

Chemical properties describe how substances change into new substances.

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.6, Part A. Then check with your teacher.

1. Physical or chemical property: iron rusts
2. Physical or chemical property: ice melts at 0°C
3. Why is gold used in electronics?

REMEMBER

LESSON 2.7
Week 4 · Lesson 1 of 2

Unit 2: Properties and Changes of Matter

Lesson 2.7: Physical and Chemical Changes

TODAY'S GOAL

I can tell physical changes from chemical changes using evidence.

Standards: MS-PS1-2

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.7 in this book with your teacher.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.7, Part A.
- [ ] Complete the rest of Workbook Lesson 2.7 (Parts B–E).

1 Get Ready

I WONDER

Is toasting bread the same kind of change as cutting it? How could you prove it?

2 Learn

In a physical change, a substance changes size, shape, or state, but it's still the same substance. Cutting, folding, crushing, melting, freezing, and dissolving are physical changes.

In a chemical change, or chemical reaction, the atoms of the starting substances, the reactants, rearrange to form new substances, the products, with different properties. Burning, rusting, cooking an egg, and baking a cake are chemical changes.

WORDS & MORE

WORDS TO KNOW

physical change
A change in size, shape, or state that doesn't make a new substance.

chemical change
A change that makes one or more new substances (a chemical reaction).

reactant
A substance present at the start of a chemical reaction.

product
A new substance made by a chemical reaction.

precipitate
A solid that forms when two liquids react.

REMEMBER

Evidence of a chemical change	Example
gas bubbles form (not from boiling)	baking soda and vinegar fizz
a color change	a banana browning; iron rusting
a temperature change (heat or cold)	a hand warmer heating up; a cold pack chilling
light is given off	a glow stick; a burning candle
a precipitate (solid) forms	a cloudy solid appears when two clear liquids mix
a new odor	milk turning sour

Evidence suggests a chemical change, but you need to check: boiling water bubbles, yet it's a physical change. The best test is to ask whether a new substance with new properties formed.

KEY IDEA

Physical changes alter size, shape, or state but not identity. Chemical changes produce new substances; evidence includes gas, color change, temperature change, light, precipitates, and new odors.

EXPLAIN IT · ANALYZE THE EVIDENCE

A student mixes two clear liquids. The mixture turns cloudy and a white solid settles to the bottom.

1 Observation: two clear liquids made a solid.

2 Evidence: a precipitate is a new substance with different properties.

3 Conclusion: the atoms rearranged into a new substance.

Explanation: This is a chemical change; the precipitate is a product.

✓ REMEMBER

Physical changes keep the substance; chemical changes make new substances.

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.7, Part A. Then check with your teacher.

1. Physical or chemical change: melting butter
2. Physical or chemical change: burning a match
3. Name two kinds of evidence of a chemical change.

WORDS & MORE

REMEMBER

LESSON 2.8
Week 4 · Lesson 2 of 2

Unit 2: Properties and Changes of Matter

WORDS & MORE

Lesson 2.8: Lab: Evidence of a Chemical Reaction

TODAY'S GOAL

I can plan and carry out a lab, collect evidence of a chemical reaction, and write a claim, evidence, and reasoning.

Standards: MS-PS1-2, MS-PS1-5

TODAY'S ASSIGNMENT

- [ ] Read Lesson 2.8 in this book with your teacher.
- [ ] Do the Investigate activity: Balloon in a Bottle.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.8, Part A.
- [ ] Complete the rest of Workbook Lesson 2.8 (Parts B–E).

WORDS TO KNOW

hypothesis
A testable prediction that explains what you expect and why.

independent variable
The factor you change on purpose in an experiment.

dependent variable
The factor you measure.

conservation of mass
In a chemical reaction, matter is not created or destroyed; the total mass stays the same.

1 Get Ready

I WONDER

When baking soda and vinegar fizz, does the mixture lose mass? Where would the mass go?

2 Learn

In this lab, you'll react baking soda (sodium bicarbonate) with vinegar (acetic acid). The reaction produces carbon dioxide gas, water, and sodium acetate. You'll look for

REMEMBER

evidence of a chemical change and test whether mass is conserved.

Scientists have shown that in any chemical reaction, atoms are rearranged but never created or destroyed. This is the law of conservation of mass. If you trap the gas in a closed container, the total mass before and after the reaction should be the same. In an open container, the gas escapes, so the mass you measure goes down.

WORDS & MORE

Part of the lab	What to write
question	Is mass conserved when baking soda reacts with vinegar?
hypothesis	If… then… because…
variables	independent: open or sealed container; dependent: total mass
data	mass before and after, plus observations
CER	claim, evidence from your data, reasoning about atoms

KEY IDEA

Gas, temperature change, and other evidence show a chemical reaction. Atoms are rearranged, not created or destroyed, so mass is conserved in a closed system.

INVESTIGATE · BALLOON IN A BOTTLE

Is mass conserved in a chemical reaction?

You need: a kitchen scale; a small plastic bottle; a balloon; baking soda (1 teaspoon); vinegar (about 50 mL); a funnel.

1. Pour the vinegar into the bottle. Use the funnel to put the baking soda into the balloon.
2. Stretch the balloon over the bottle opening without spilling the baking soda. Measure the total mass.
3. Lift the balloon so the baking soda falls in. Record your observations.
4. When the fizzing stops, measure the total mass again. Then repeat with the bottle open (no balloon) and compare.

Record: your hypothesis, a data table of masses and observations for both trials, and a CER paragraph.

Safety: Wear goggles. Point the bottle away from faces.

✓ REMEMBER

Evidence shows a reaction happened; mass is conserved when no matter escapes.

REMEMBER

WORDS & MORE

3 Check Your Understanding

✎ CHECK YOUR UNDERSTANDING

Directions: Answer each question. Write your answers in Workbook Lesson 2.8, Part A. Then check with your teacher.

1. What gas forms when baking soda reacts with vinegar?
2. Why does the open container lose mass?
3. What does conservation of mass mean?

UNIT 2 REVIEW

Unit 2 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
2.1	Matter is made of atoms; molecules and compounds are atoms joined together.
2.2	Physical properties identify substances; states differ in particle arrangement and motion.
2.3	Thermal energy changes particle motion and can change state; temperature holds steady during the change.
2.4	Density = mass ÷ volume; it predicts sinking and floating and identifies substances.
2.5	Mixtures can be separated by properties such as solubility and magnetism; conductivity is another key property.
2.6	Chemical properties describe how substances change into new substances.
2.7	Physical changes keep the substance; chemical changes make new substances.
2.8	Evidence shows a reaction happened; mass is conserved when no matter escapes.

Glossary

atmosphere	The layer of gases around Earth.
atom	The smallest particle of an element.
biodiversity	The variety of life in an area.
biosphere	All living things on Earth.
buoyancy	The upward push of a liquid on an object in it.
carbon cycle	The movement of carbon among the air, living things, oceans, and rocks.
cellular respiration	Glucose + oxygen → CO_2 + water + energy.
chemical change	A change that makes new substances.
chemical property	A property that describes how a substance can change into a new substance.
chlorophyll	The green pigment that captures light energy.
chloroplast	The part of a plant cell where photosynthesis happens.
combustion	Burning; a reaction with oxygen that releases energy and carbon dioxide.
compound	Different elements chemically joined.
condensation	The change from water vapor to liquid water.
conductivity	How well a material lets heat or electricity pass through it.
conservation of mass	Mass is not created or destroyed in a reaction.
consumer	An organism that eats other organisms.
controlled variables	Conditions kept the same so the test is fair.
decomposer	An organism that breaks down dead matter and returns materials to the environment.
density	Mass per volume.
dependent variable	The variable that is measured.
displacement	Measuring volume by how much water an object pushes aside.
ecosystem services	Benefits people get from healthy ecosystems, such as clean water and pollination.
element	A pure substance made of only one kind of atom.
evaporation	The change from liquid to gas at the surface of a liquid.
fermentation	A process that releases energy from sugar without oxygen.
flammability	How easily a substance burns.

food web	A diagram of all the feeding relationships in an ecosystem.
fossil fuel	Coal, oil, or natural gas formed from ancient living things.
geosphere	The solid rock and soil of Earth.
glucose	A sugar made in photosynthesis that stores energy.
hydrosphere	All of Earth's water.
hypothesis	A testable prediction that explains how one variable affects another.
independent variable	The variable that is changed.
legume	A plant such as a bean, pea, or peanut that hosts nitrogen-fixing bacteria in its roots.
malleable	Able to be hammered or pressed into shapes without breaking.
mass	The amount of matter in an object.
matter	Anything that has mass and takes up space.
melting point	The temperature at which a solid becomes a liquid.
mitochondria	Cell parts where cellular respiration releases energy.
mixture	Two or more substances mixed together but not chemically joined.
molecule	Two or more atoms joined together.
nitrogen fixation	Changing N_2 into usable forms.
oxidation	A reaction with oxygen, such as rusting or burning.
photosynthesis	Light + CO_2 + water → glucose + oxygen.
physical change	A change in size, shape, or state that doesn't make a new substance.
physical property	A property you can observe or measure without changing what the substance is.
pollinator	An animal, such as a bee, that carries pollen between flowers.
precipitate	A solid that forms when two liquids react.
producer	An organism that makes its own food, usually by photosynthesis.
product	A new substance made by a chemical reaction.
reactant	A substance present at the start of a chemical reaction.
reactivity	How easily a substance reacts with other substances.
solubility	How much of a substance can dissolve in another.
solute	The substance that dissolves in a solution.
solution	A mixture in which one substance is spread evenly through another.

state of matter	The form matter takes: solid, liquid, or gas.
sublimation	The change from solid directly to gas.
thermal energy	The energy of moving particles.
transpiration	The release of water vapor from plant leaves.
trophic level	A feeding level in a food chain or energy pyramid.
volume	The amount of space something takes up.

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